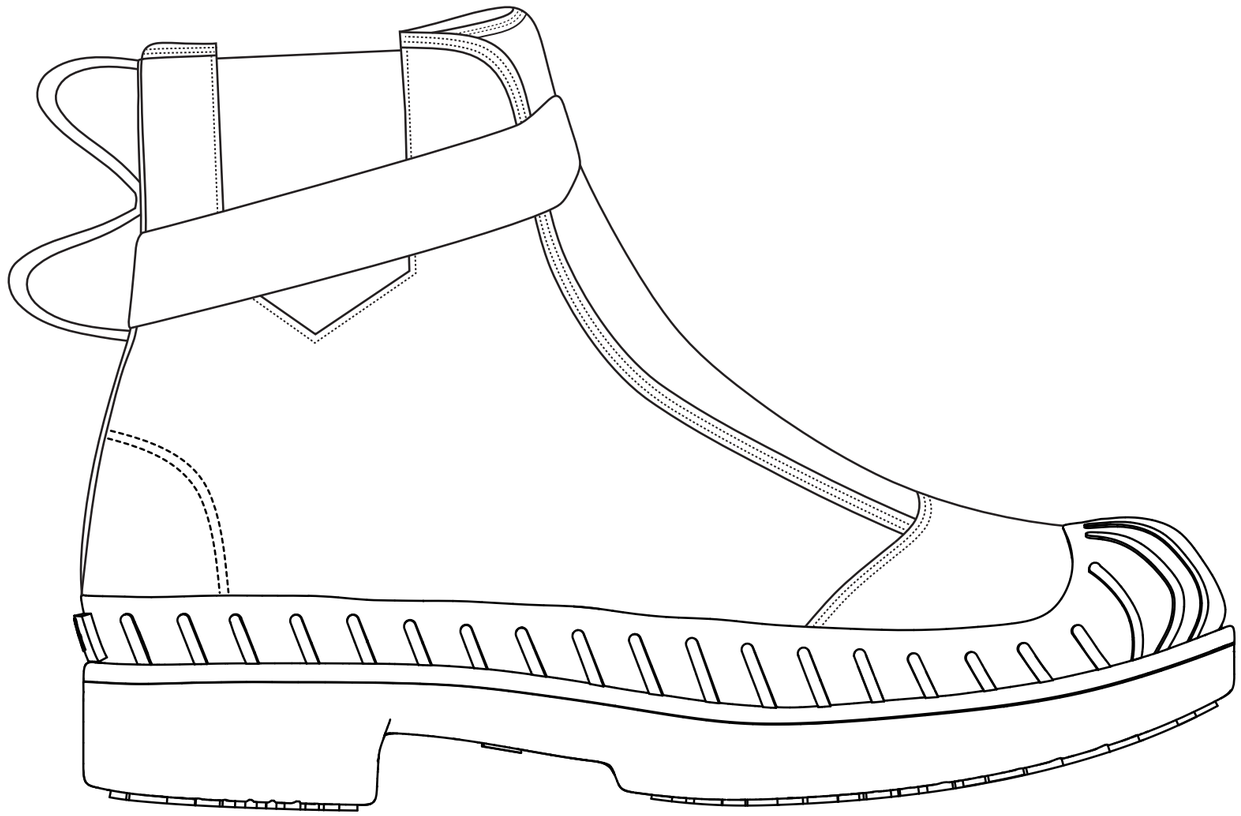
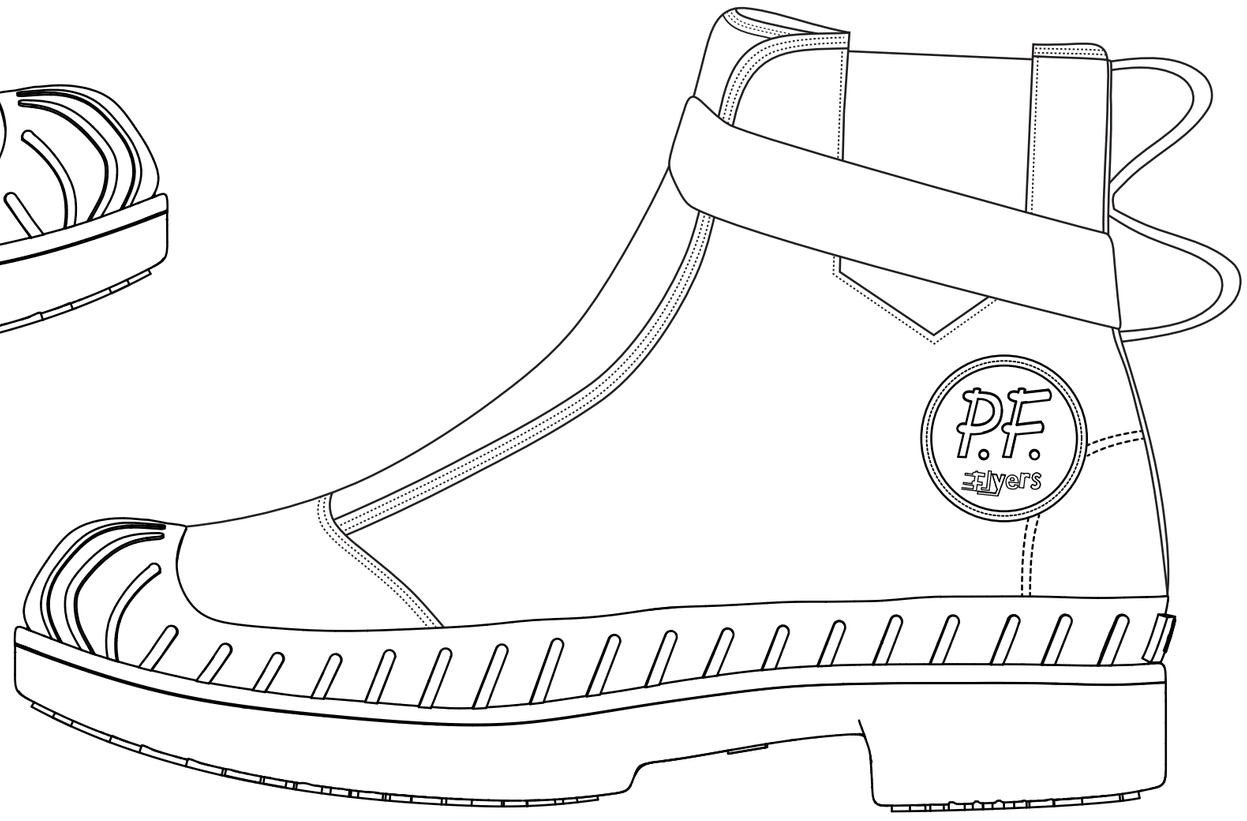




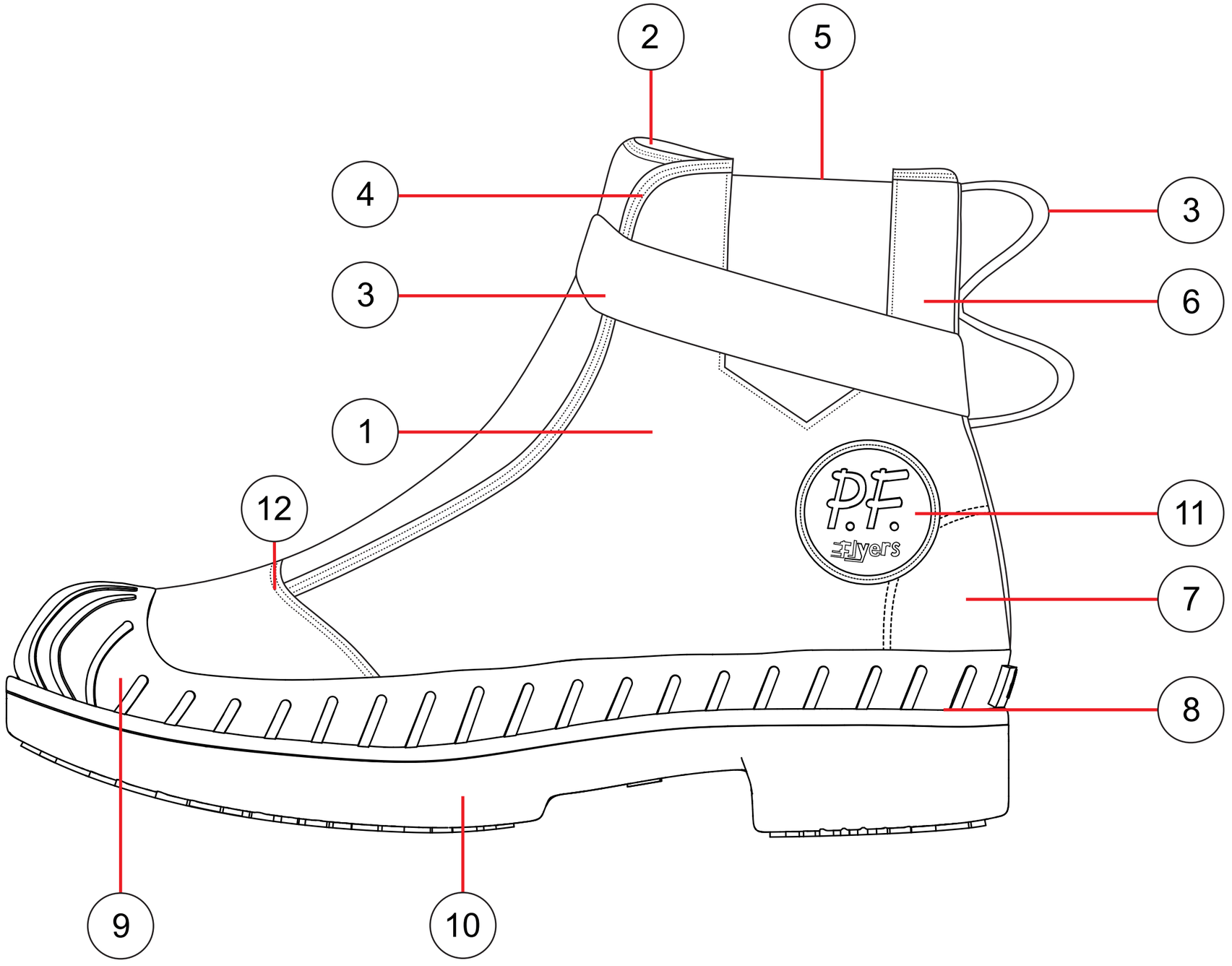
	ELLIOT HILDNER	PF CLOVER	DATE 6/9/26	UNITS mm
	SPORTS PRODUCT DESIGN	BEEKEEPING BOOT	SCALE 1:2	PAGE 1

MAT. #	ITEM	IMAGE	DESCRIPTION	SIZE	SOURCED FROM
1	UPPER		100% Organic Cotton Canvas, Coated with Aluminum Oxide + Lanolin DWR Woven	4 sq ft	UO PDX Campus
2	INTERIOR LINING		100% Cotton Woven	2 sq ft	UO PDX Campus
3	CINCH STRAP + PULL TABS		100% Cotton Strap Knit	4 ft	UO PDX Campus
4	EXTERIOR LINING		100% Cotton Bias Tape Woven	3 ft	UO PDX Campus
5	GUSSET		Elastic Gore Knit	1 sq ft	UO PDX Campus
6	PADDING		Open-Cell Foam Non-Woven	2 sq ft	UO PDX Campus

ELLIOT HILDNER	PF CLOVER	DATE 6/9/26	UNITS	
SPORTS PRODUCT DESIGN	BOM	SCALE	PAGE 2	

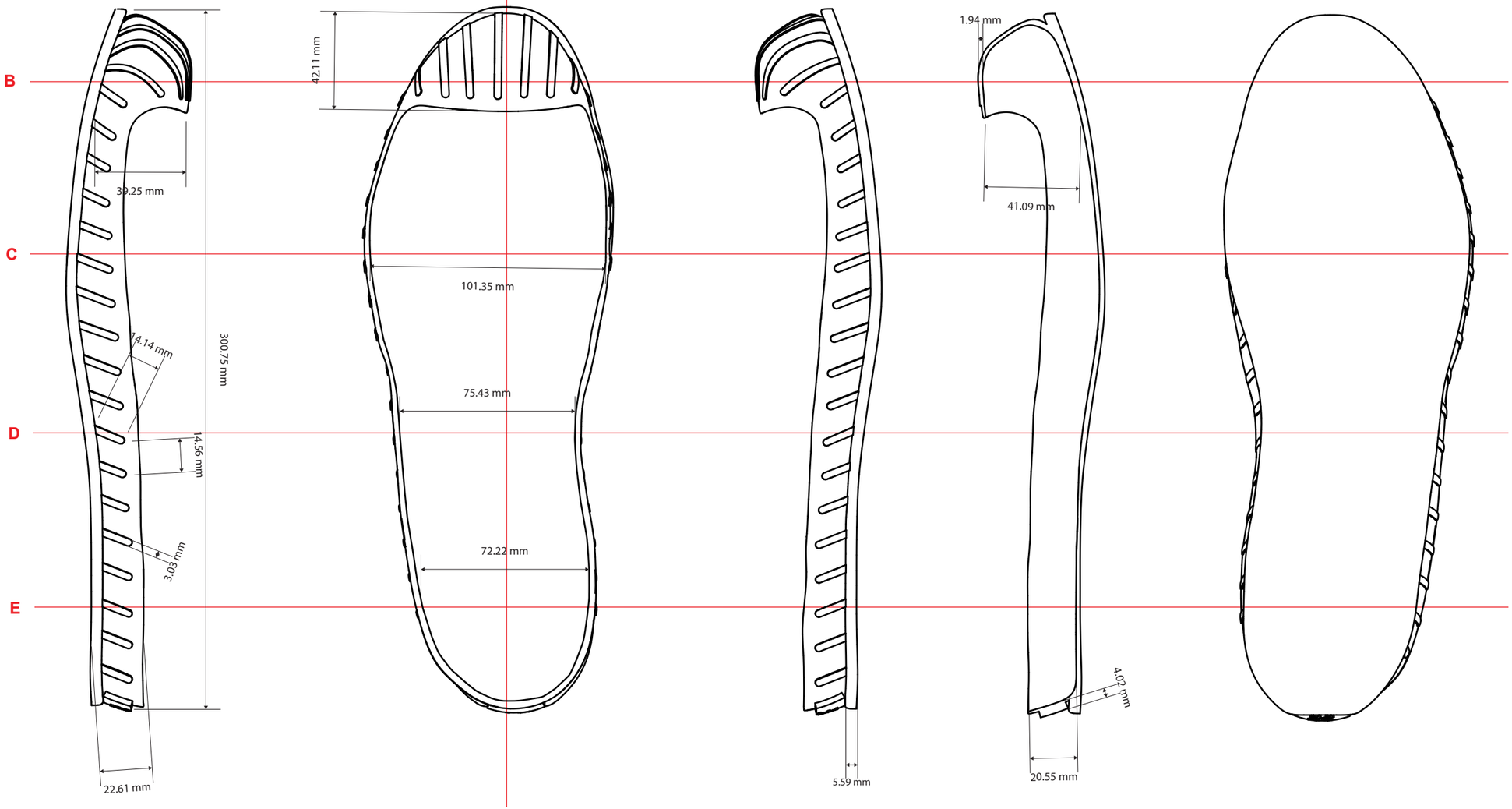
MAT. #	ITEM	IMAGE	DESCRIPTION	SIZE	SOURCED FROM
7	HEEL COUNTER		Leather	0.25 sq ft	UO PDX Campus
8	STROBEL		100% Polyester Non-Woven	1 sq ft	UO PDX Campus
9	MIDSOLE		TPU	1 filament roll	UO PDX Campus
10	OUTSOLE		TPU	1 filament roll	UO PDX Campus
11	LOGO PATCH		Leather	0.25 sq ft	UO PDX Campus
12	THREAD		100% Cotton	1 spool	Coats & Clark

ELLIOT HILDNER	PF CLOVER	DATE 6/9/26	UNITS	
SPORTS PRODUCT DESIGN	BOM	SCALE	PAGE 3	



	ELLIOT HILDNER	PF CLOVER	DATE 6/9/26	UNITS mm
	SPORTS PRODUCT DESIGN	CALL-OUTS	SCALE 1:2	PAGE 4

LATERAL VIEW



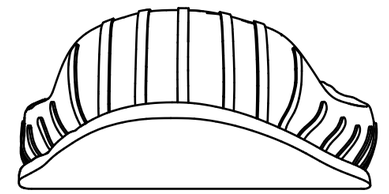
TOP VIEW

MEDIAL VIEW

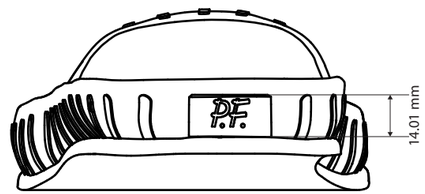
SECTION A-A

BOTTOM VIEW

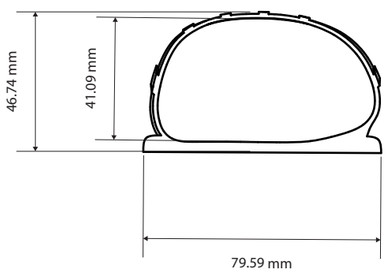
TOP VIEW



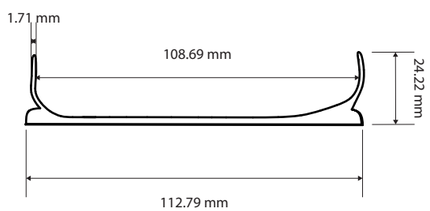
BACK VIEW



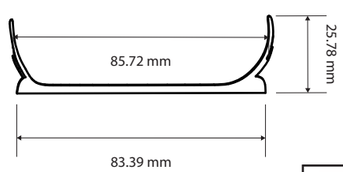
SECTION B-B



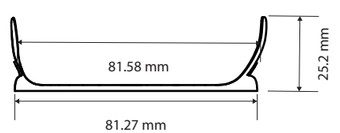
SECTION C-C



SECTION D-D



SECTION E-E

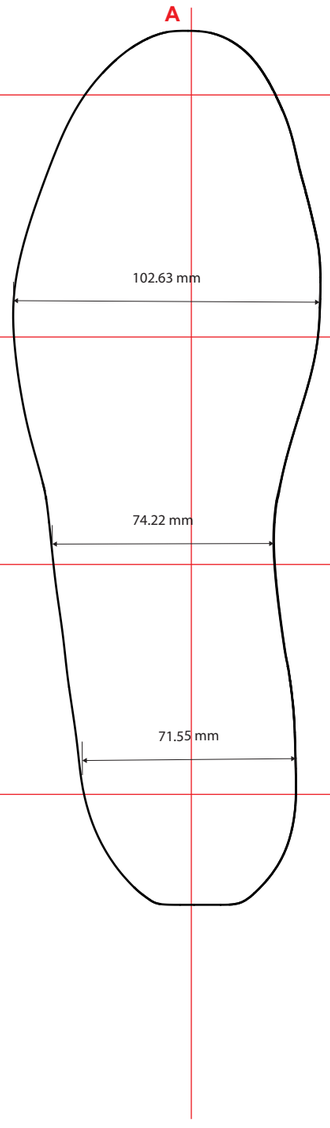


	ELLIOT HILDNER	PF CLOVER	DATE 6/9/26	UNITS mm
	SPORTS PRODUCT DESIGN	MIDSOLE	SCALE 1:2	PAGE 5

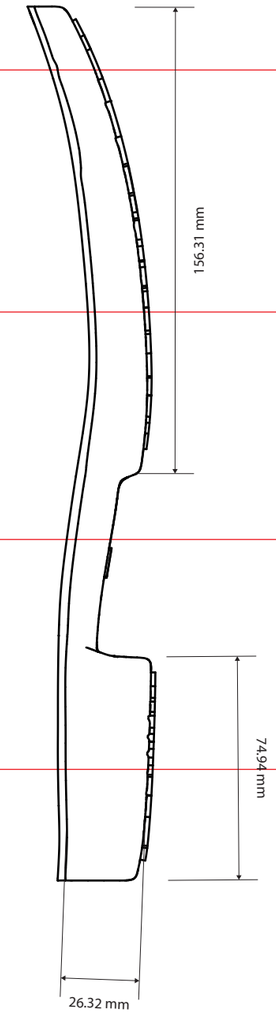
LATERAL VIEW



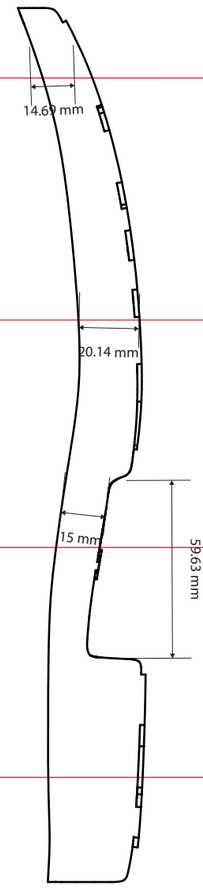
TOP VIEW



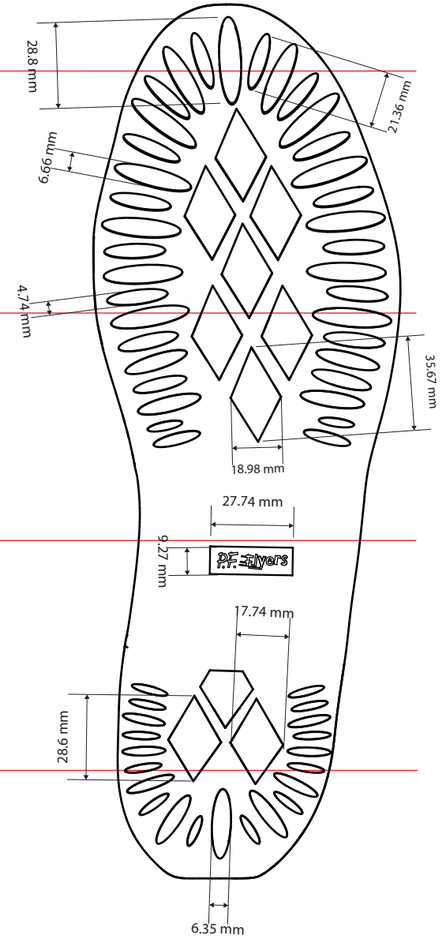
MEDIAL VIEW



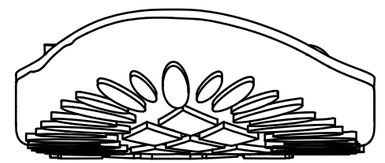
SECTION A-A



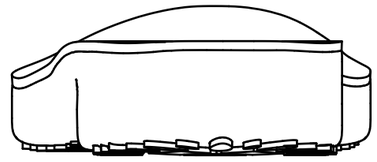
BOTTOM VIEW



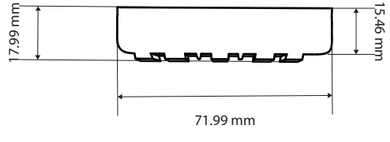
TOP VIEW



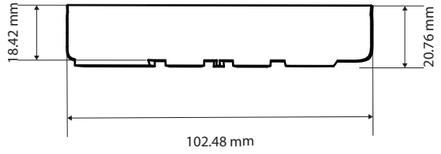
BACK VIEW



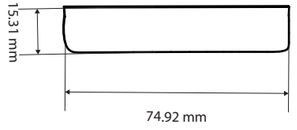
SECTION B-B



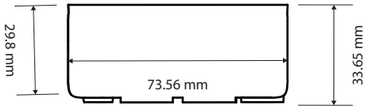
SECTION C-C



SECTION D-D



SECTION E-E



	ELLIOT HILDNER	PF CLOVER	DATE 6/9/26	UNITS mm
	SPORTS PRODUCT DESIGN	OUTSOLE	SCALE 1:2	PAGE 6

= Grain Line

Strobel
Cut 1

Tongue
Cut 1

Tongue Lining
Cut 1

Medial Side
Cut 1

Medial Lining
Cut 1

Lateral Side
Cut 1

Lateral Lining
Cut 1

Gusset
Cut 2

Forefoot
Cut 1

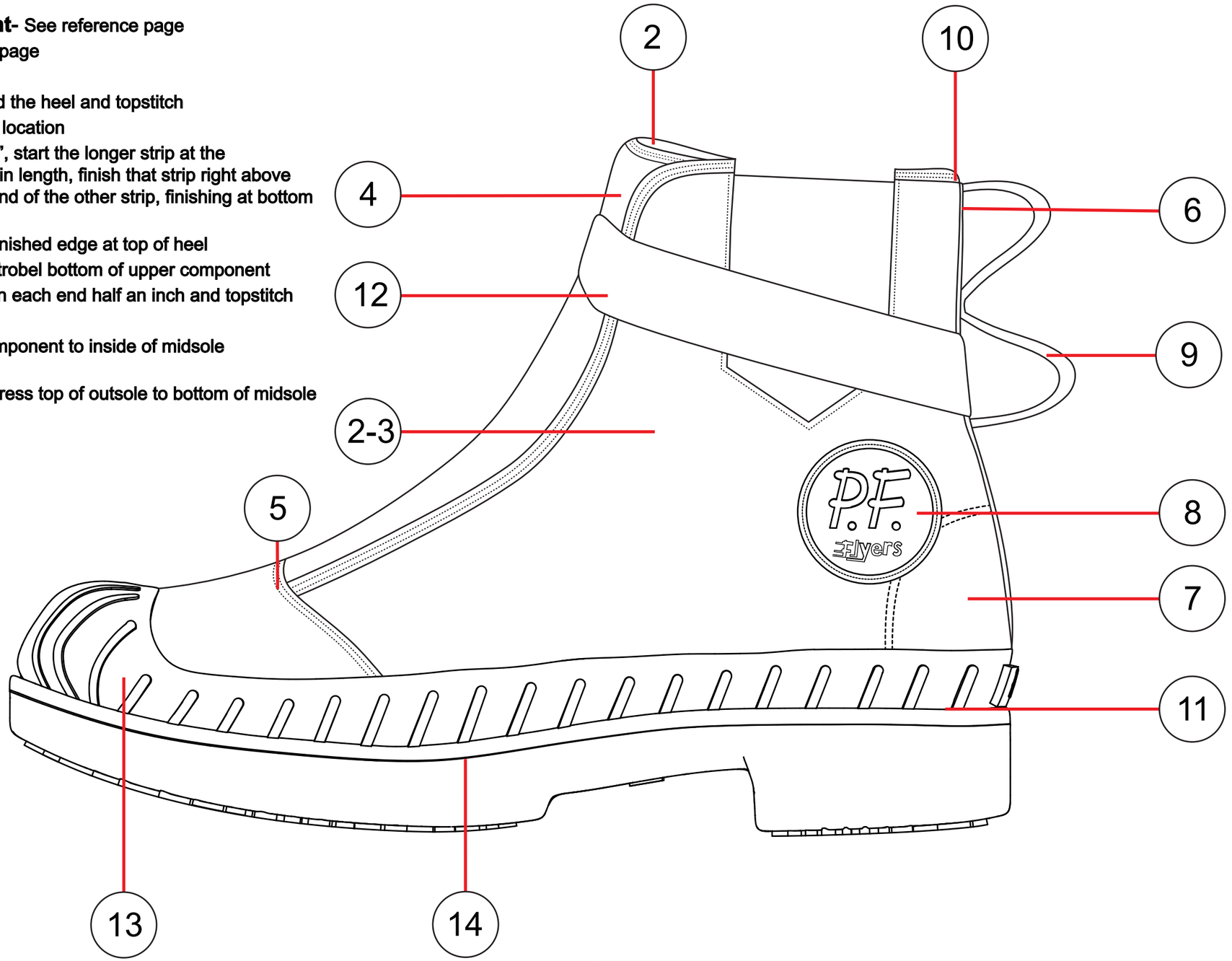
Heel Counter
Cut 1

Padding
Cut 2

Logo Patch
Cut 1

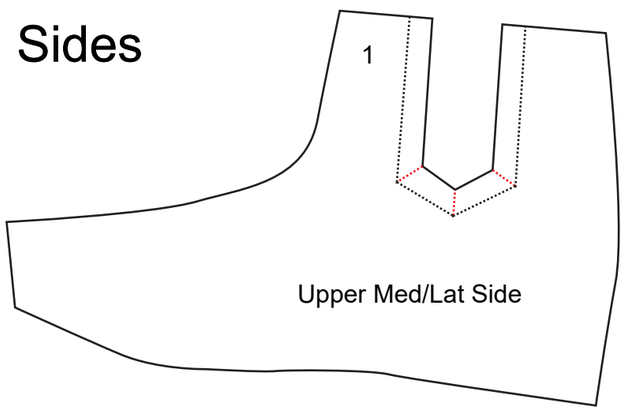
	ELLIOT HILDNER	PF CLOVER	DATE 6/9/26	UNITS
	SPORTS PRODUCT DESIGN	PATTERN	SCALE 1:2	PAGE 7

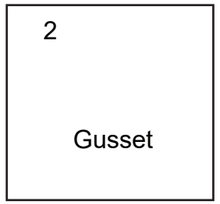
1. **Cut pattern pieces** - Cut all upper components
2. **Assemble Medial & Lateral Components** - See reference page
3. **Assemble Tongue Component** - See reference page
4. **Sew Tongue Component to each Side Component**- See reference page
5. **Sew Forefoot to Upper Component** - See reference page
6. **Sew Sides together at Heel** - Using a zigzag stitch
7. **Sew in Heel Counter** - Place at specified location around the heel and topstitch
8. **Sew on Logo Patch**- Topstitch leather patch at specified location
9. **Sew on Pull Tabs** - Cut two strips, one at 8" and one at 5", start the longer strip at the top of the heel, sew it in, now make two pull tabs, both 1-3/4" in length, finish that strip right above the heel counter. Sew the shorter strip starting on top of the end of the other strip, finishing at bottom of upper
10. **Apply Lining to Heel Edge**- Put bias tape around unfinished edge at top of heel
11. **Strobel Upper Component**- Use strobel machine to strobel bottom of upper component
12. **Assemble Cinch Strap** - Cut Cinch to 18 inches, fold in each end half an inch and topstitch to finish, apply velcro
13. **Adhesive Glue Upper to Midsole** - Press upper component to inside of midsole with adhesive
14. **Adhesive Glue Outsole to bottom of Midsole** - Press top of outsole to bottom of midsole and add adhesive

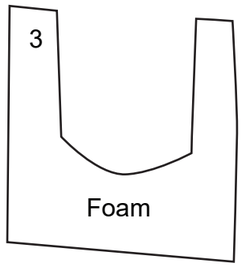


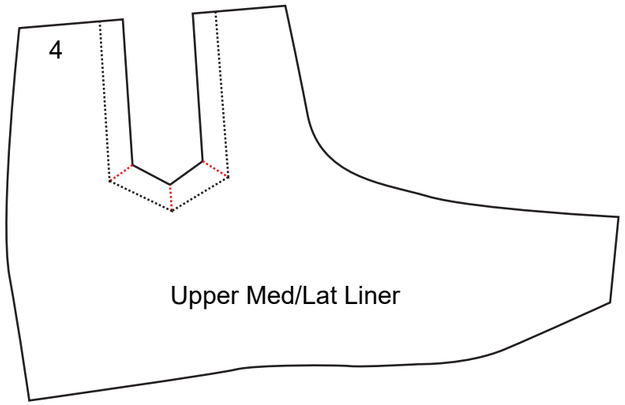
	ELLIOT HILDNER	PF CLOVER	DATE 6/9/26	UNITS
	SPORTS PRODUCT DESIGN	ORDER OF OPERATIONS	SCALE	PAGE 8

Sides

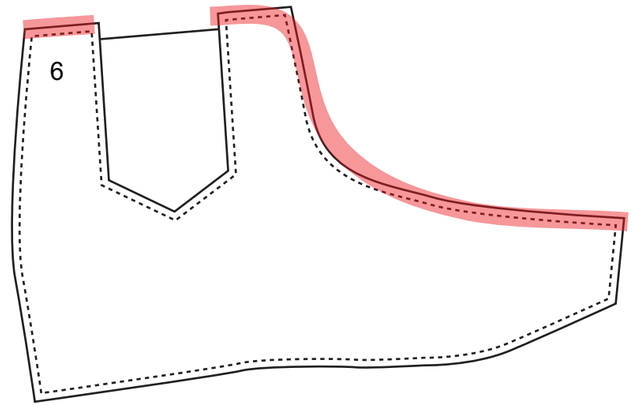






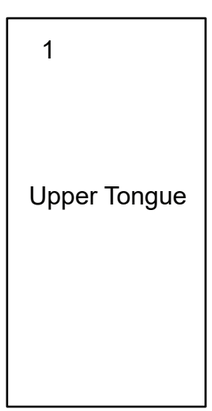




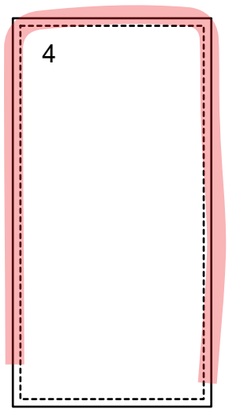


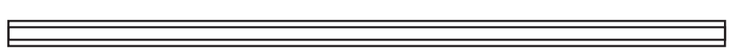
1. **Notch Out Gusset Area -** On pieces 1 & 2, allowing 1/2" SA, red dotted lines are cuts, black are fold
2. **Fold in SA** - Fold inward to WRONG SIDE along the black dotted lines
3. **Layering Pieces** - Place piece 1 WRONG SIDE up, then piece 2, then piece 3, then piece 4 WRONG SIDE down.
4. **Sew Perimeter**- Topstitch around the perimeter of the layered pieces, allowing 1/4" SA (becomes piece 6)
5. **Sew Bias Tape** - Apply bias tape (piece 5) along highlighted area to finish component (piece 6), allowing 3/8" over on the front, topstitch

Tongue





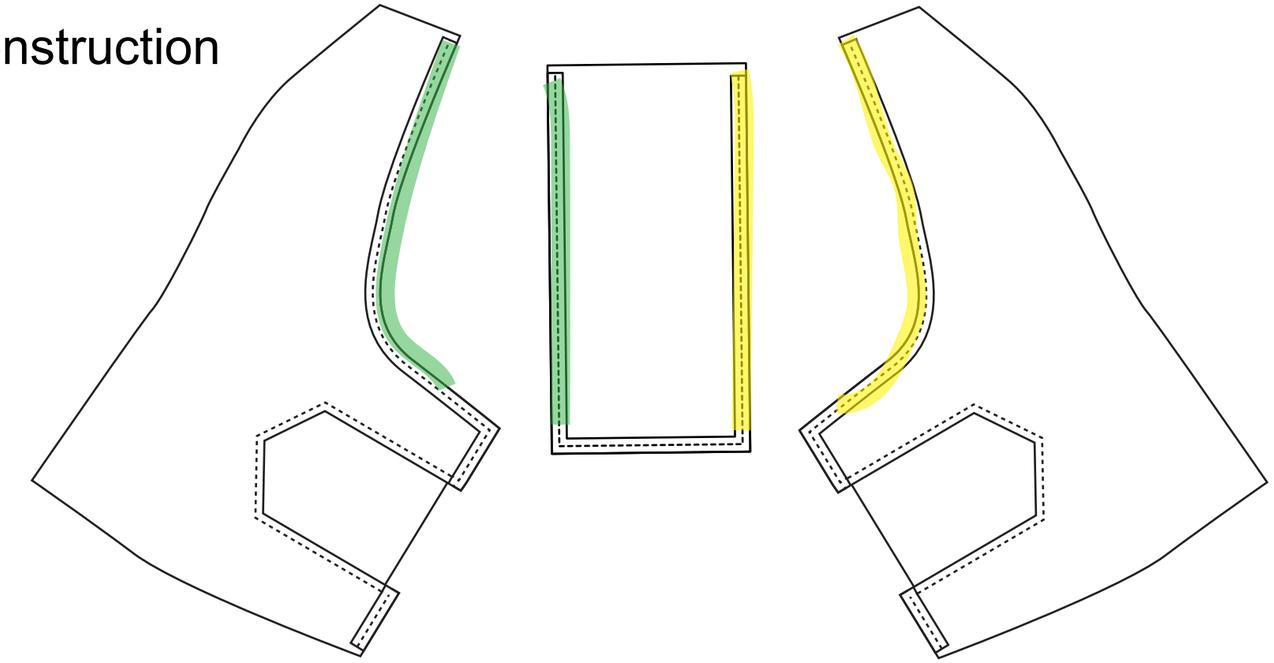


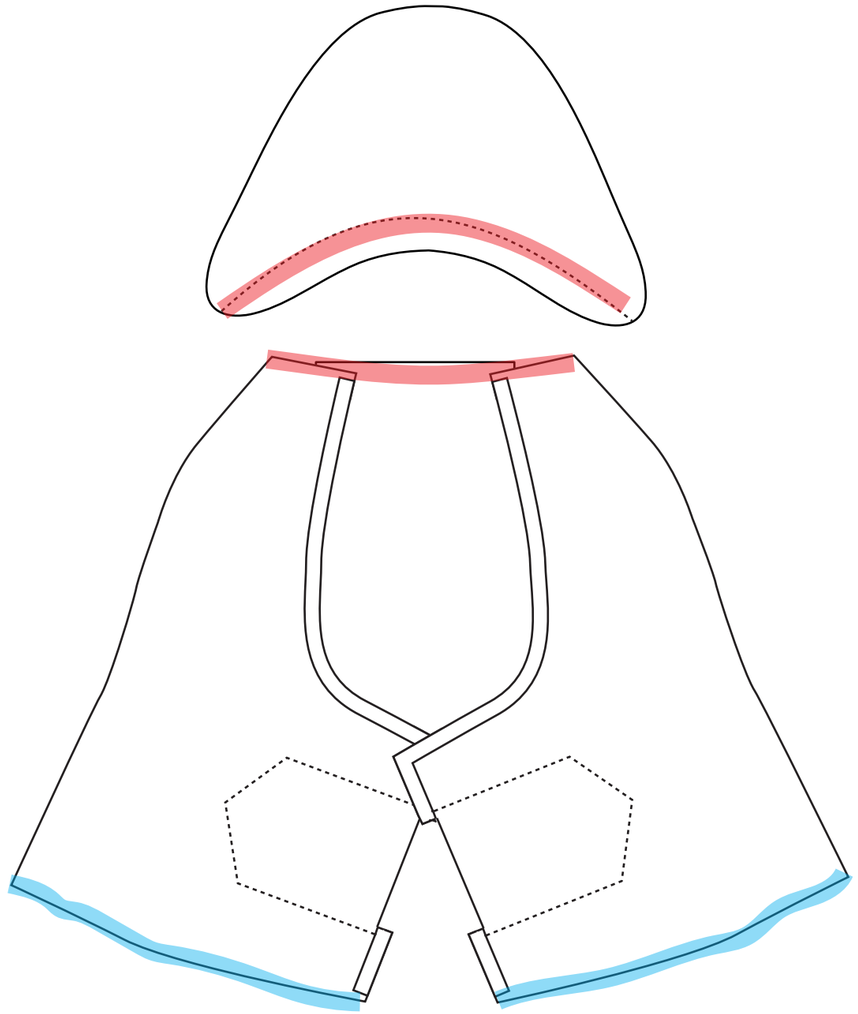


1. **Layer Pieces 1 & 2 -** Place piece 1 WRONG SIDE up, place piece 2 ontop of it WRONG SIDE down
2. **Sew Perimeter** - Topstitch perimiter with 1/4" SA (becomes piece 4)
3. **Apply Bias Tape** - Apply bias tape (piece 3) around highlighted part of piece 4
4. **Sew Perimeter**- Topstitch around the perimeter of the layered pieces, allowing 1/4" SA

	ELLIOT HILDNER	PF CLOVER	DATE 6/9/26	UNITS
	SPORTS PRODUCT DESIGN	CONSTRUCTION DETAILS	SCALE 1:2	PAGE 9

Upper Construction

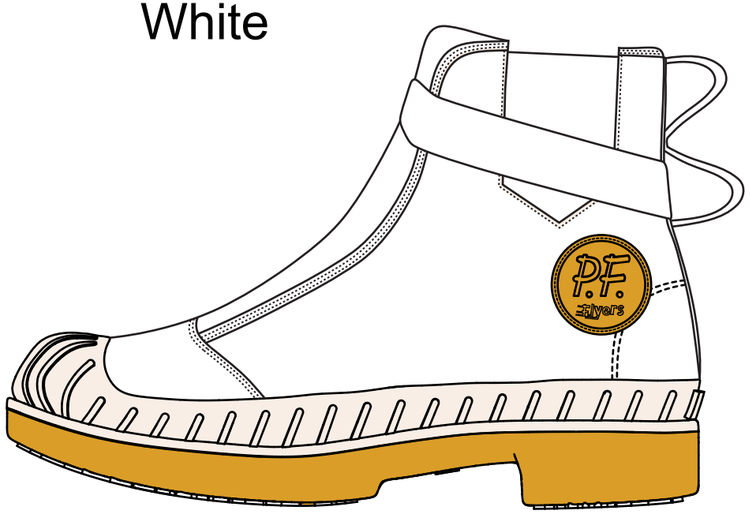




1. **Sew Lateral Side to Tongue -** Sew along green highlighted lines, topstitch
2. **Sew Medial Side to Tongue** - Sew along yellow highlighted lines, topstitch
3. **Sew Forefoot to Upper** - Sew along red highlighted lines, topstitch
4. **Sew Heel Together**- Sew along blue highlighted lines, zigzag stitch

	ELLIOT HILDNER	PF CLOVER	DATE 6/9/26	UNITS
	SPORTS PRODUCT DESIGN	UPPER CONSTRUCTION	SCALE	PAGE 10

Beekeeping Suit
White



hex #
ffffff

hex #
f9eee5

hex #
da9d28

Honeycomb
Yellow

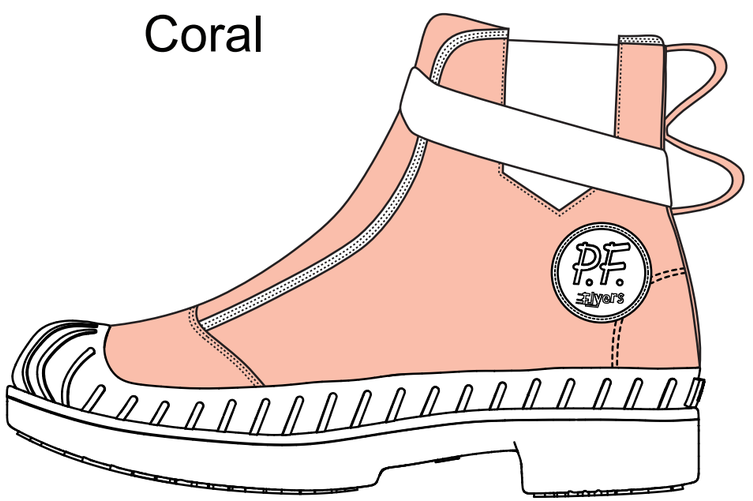


hex #
fff200

hex #
ffffff

hex #
ffffff

Floral
Coral



hex #
fabeab

hex #
ffffff

hex #
ffffff

	ELLIOT HILDNER	PF CLOVER	DATE 6/9/26	UNITS
	SPORTS PRODUCT DESIGN	COLORWAYS	SCALE 1:2	PAGE 11